

Observer-Based Nonlinear Control of Space Vehicles with Multi-Mass Fuel Slosh Dynamics

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Abstract—This paper studies the problem of observer-based control of space vehicles with fuel slosh dynamics in a zero gravity environment. Multi-mass-spring models are considered for the characterization of the most prominent sloshing modes. The control objective is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes. A full-state feedback that uses a reduced-order observer for the estimation of the slosh states is proposed to achieve the objective. The effectiveness of the proposed observer-based control law is illustrated through a computer simulation.

I. INTRODUCTION

There is an extensive body of literature on the interaction of vehicle dynamics and slosh dynamics and their control. It has been demonstrated that pendulum and mass-spring models can approximate complicated fluid and structural dynamics; such models have formed the basis for many studies on dynamics and control of space vehicles with fuel slosh [18]. Some of the works include a single pendulum model [6] or a single mass-spring model [24] in the dynamics formulation to represent the first fluid slosh mode. The literature that incorporates more than one sloshing mode in modeling the slosh dynamics includes [7], [10], [20], [21], [22], [23], [25].

The earlier control approaches developed for accelerating space vehicles and robotic systems moving liquid-filled containers have commonly employed methods of linear control design ([2], [4], [8], [11], [12], [26], [27], [29], [30], [31], [32]). The linear control design methods that use command input shaping methods do not require sensor measurements of the sloshing liquid ([1], [8], [17], [19], [32]). The control methods that require either sensor measurements ([11], [12]) or observer estimates of the slosh states [3] have recently received an increased attention.

In this paper, we consider the problem of observer-based nonlinear control of slosh in planar maneuvering space vehicles in a zero gravity environment. A full-state feedback that uses a reduced-order observer for the estimation of the slosh state is proposed for the control of the translational velocity vector and the attitude of the spacecraft while suppressing the sloshing modes. The effectiveness of the proposed observer-

based control law is demonstrated through a computer simulation.

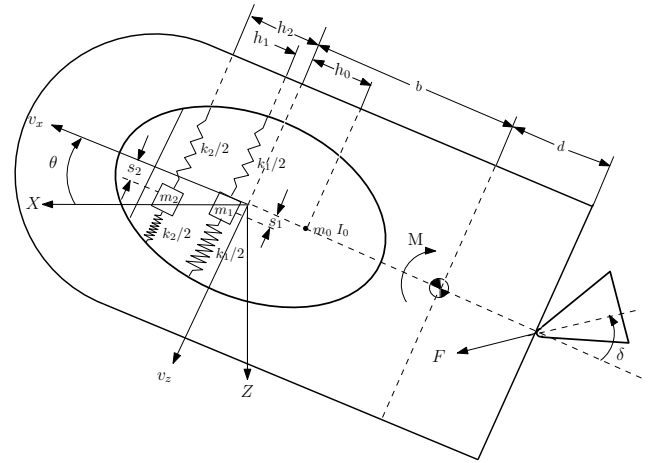


Figure 1. A multiple slosh mass-spring model for a spacecraft [21].

II. MATHEMATICAL MODEL

Following the development in [21], consider a rigid spacecraft moving on a plane as indicated in Fig. 1, where v_x , v_z are the axial and transverse components, respectively, of the velocity of the center of the fuel tank, and θ denotes the attitude angle of the spacecraft with respect to a fixed reference. The fluid is modeled by moment of inertia I_0 assigned to a rigidly attached mass m_0 and point masses m_i , $i = 1, \dots, N$, whose relative positions along the spacecraft fixed z -axis are denoted by s_i . The locations $h_0 > 0$ and $h_i > 0$ are referenced to the center of the tank. A restoring force $-k_i s_i$ acts on the mass m_i whenever the mass is displaced from its neutral position $s_i = 0$. A thrust F is produced by a gimballed thrust engine as shown in Fig. 1, where δ denotes the gimbal deflection angle, which is considered as one of the control inputs. A pitching moment M is also available for control purposes. The constants in the problem are the spacecraft mass m and

moment of inertia I ; the fuel masses m_0, m_i ; the distance b between the body z -axis and the spacecraft center of mass location along the longitudinal axis, and the distance d from the gimbal pivot to the spacecraft center of mass. If the tank center is in front of the spacecraft center of mass then $b > 0$. The parameters m_i, h_i, k_i and b depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank.

The equations of motion are given by

$$(m + m_f)(\dot{v}_x + \dot{\theta}v_z) + mb\dot{\theta}^2 + \sum_{i=1}^N m_i(s_i\ddot{\theta} + 2\dot{s}_i\dot{\theta}) = F \cos \delta, \quad (1)$$

$$(m + m_f)(\dot{v}_z - \dot{\theta}v_x) + mb\ddot{\theta} + \sum_{i=1}^N m_i(\ddot{s}_i - s_i\dot{\theta}^2) = F \sin \delta, \quad (2)$$

$$I_t\ddot{\theta} + \sum_{i=1}^N m_i(s_i(\dot{v}_x + \dot{\theta}v_z) - h_i\ddot{s}_i + 2s_i\dot{s}_i\dot{\theta}) + mb(\dot{v}_z - \dot{\theta}v_x) = M + F(b + d) \sin \delta, \quad (3)$$

$$m_i(\ddot{s}_i + \dot{v}_z - \dot{\theta}v_x - h_i\ddot{\theta} - s_i\dot{\theta}^2) + k_i s_i + c_i \dot{s}_i = 0, \quad \forall i, \quad (4)$$

where c_i are the damping constants representing the dissipative effects due to fuel slosh and

$$I_t = I + I_0 + mb^2 + m_0 h_0^2 + \sum_{i=1}^N m_i(h_i^2 + s_i^2).$$

The control objective is to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while suppressing the fuel slosh modes.

III. FEEDBACK CONTROL LAW

In this section, we summarize our nonlinear control design that appeared in [21]. Consider the model of a spacecraft with a gimbaled thrust engine shown in Fig. 1. If the thrust F during the fuel burn is a positive constant, and if the gimbal deflection angle and pitching moment are zero, $\delta = M = 0$, then the spacecraft and fuel slosh dynamics have a relative equilibrium defined by

$$v_z = \bar{v}_z, \quad \theta = \bar{\theta}, \quad \dot{\theta} = 0, \quad s_i = 0, \quad \dot{s}_i = 0, \quad \forall i,$$

where $\bar{v}_z$ and $\bar{\theta}$ are arbitrary constants. Without loss of generality in our subsequent analysis, we consider the relative equilibrium at the origin, i.e. $\bar{v}_z = 0, \bar{\theta} = 0$. Note that the relative equilibrium corresponds to the vehicle axial velocity

$$v_x(t) = \frac{F}{m + m_f}t + v_{x0}, \quad t \leq t_b,$$

where v_{x0} is the initial axial velocity of the spacecraft and t_b is the fuel burn time.

Now assume the axial acceleration $a_x = \dot{v}_x + \dot{\theta}v_z$ is not significantly affected by small gimbal deflections, pitch

changes and fuel motion (an assumption verified in simulations). Consequently, equation (1) becomes:

$$\dot{v}_x + \dot{\theta}v_z = \frac{F}{m + m_f} = \bar{a}_x. \quad (5)$$

Substituting this approximation leads to the following equations of motion for the transverse, pitch and slosh dynamics:

$$(m + m_f)(\dot{v}_z - \dot{\theta}v_x(t)) + mb\ddot{\theta} + \sum_{i=1}^N m_i(\ddot{s}_i - s_i\dot{\theta}^2) = F \sin \delta, \quad (6)$$

$$I_t\ddot{\theta} + \sum_{i=1}^N m_i[\bar{a}_x s_i - h_i\ddot{s}_i + 2s_i\dot{s}_i\dot{\theta}] + mb(\dot{v}_z - \dot{\theta}v_x) = M + F(b + d) \sin \delta, \quad (7)$$

$$\ddot{s}_i + \dot{v}_z - \dot{\theta}v_x(t) - h_i\ddot{\theta} - s_i\dot{\theta}^2 + \omega_i^2 s_i + 2\zeta_i \omega_i \dot{s}_i = 0, \quad \forall i, \quad (8)$$

where $v_x(t)$ is considered as an exogenous input and

$$\omega_i^2 = \frac{k_i}{m_i}, \quad 2\zeta_i \omega_i = \frac{c_i}{m_i}, \quad \forall i.$$

Here ω_i and $\zeta_i, \forall i$, denote the undamped natural frequencies and damping ratios, respectively. The subsequent analysis is based on the above equations of motion for the transverse, pitch and slosh dynamics of the vehicle.

To design a nonlinear controller that stabilizes the relative equilibrium at the origin of the equations (6)-(8), define control transformations from (δ, M) to new control inputs (u_1, u_2) so that the equations (6)-(8) can be written as:

$$\dot{v}_z = u_1 + \dot{\theta}v_x(t), \quad (9)$$

$$\ddot{\theta} = u_2, \quad (10)$$

$$\ddot{s}_i = -\omega_i^2 s_i - 2\zeta_i \omega_i \dot{s}_i - u_1 + h_i u_2 + s_i \dot{\theta}^2, \quad \forall i. \quad (11)$$

In [21], it is shown that the feedback laws

$$u_1 = -l_1[r_1 v_z - r_4 \sum_{i=1}^N (\dot{s}_i - h_i \dot{\theta})], \quad (12)$$

$$u_2 = -\frac{1}{\mu}[r_2 \theta + l_2 \dot{\theta} + r_1 v_x(t) v_z + r_4 \sum_{i=1}^N (h_i \omega_i^2 s_i + 2\zeta_i \omega_i h_i \dot{s}_i + s_i \dot{s}_i \dot{\theta} - h_i s_i \dot{\theta}^2)], \quad (13)$$

where $\mu = r_3 - r_4 \sum_{i=1}^N h_i^2 > 0$ and $r_j, j = 1, \dots, 4, l_k, k = 1, 2$, are positive control parameters, asymptotically stabilize the origin of the closed loop defined by the equations (9)-(11) and the feedback control laws (12)-(13). Note that the control parameters can be chosen arbitrarily to achieve good closed loop responses.

IV. SLOSH STATE OBSERVER DESIGN

In this section, a reduced-order slosh state observer is designed to estimate the unmeasurable slosh states. For simplicity in presenting the main ideas, we consider the $N = 2$ case.

Consider the system (9)-(11), and let

$$\mathbf{u} = \mathbf{A}_1 \mathbf{s} + \mathbf{A}_2 \dot{\mathbf{s}} + \mathbf{v},$$

where $\mathbf{u} = [u_1, u_2]^T$, $\mathbf{s} = [s_1, s_2]^T$, and $\mathbf{v} = [v_1, v_2]^T$ is the new input and

$$\mathbf{A}_1 = \begin{bmatrix} 0 & 0 \\ a_1 & a_2 \end{bmatrix},$$

$$\mathbf{A}_2 = \begin{bmatrix} 0 & 0 \\ a_3 & a_4 \end{bmatrix}.$$

Let $\mathbf{x}_r = [\theta, v_z, \dot{\theta}]^T$ and $\mathbf{x}_s = [s^T, \dot{s}^T]^T$ be the spacecraft and slosh state vectors, respectively, and denote by $\mathbf{x} = [\mathbf{x}_r^T, \mathbf{x}_s^T]^T$ the overall state vector. Let $\hat{\mathbf{x}}_s$ denote the estimated slosh state. Equations of controlled motion can then be written in state space form as

$$\dot{\mathbf{x}}_r = \mathbf{A}_r \mathbf{x}_r + \mathbf{A}_s \mathbf{x}_s + \mathbf{B}_r \hat{\mathbf{v}} + \mathbf{H}(\dot{\theta}, t), \quad (14)$$

$$\dot{\mathbf{x}}_s = \mathbf{f}_s(\dot{\theta}, \mathbf{x}_s, t) + \mathbf{g}_s \hat{\mathbf{v}}, \quad (15)$$

where $\hat{\mathbf{v}} = \hat{\mathbf{v}}(\mathbf{x}_r, \hat{\mathbf{x}}_s)$ and

$$\mathbf{A}_r = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

$$\mathbf{A}_s = \begin{bmatrix} 0 & 0 \\ \mathbf{A}_1 & \mathbf{A}_2 \end{bmatrix},$$

$$\mathbf{f}_s(\dot{\theta}, \mathbf{x}_s, t) = \begin{bmatrix} \dot{\mathbf{s}} \\ \mathbf{J}(\mathbf{A}_1 \mathbf{s} + \mathbf{A}_2 \dot{\mathbf{s}}) + \mathbf{R}(\dot{\theta}, \mathbf{s}, \dot{\mathbf{s}}) \end{bmatrix},$$

$$\mathbf{H}(\dot{\theta}, t) = \begin{bmatrix} 0 \\ \mathbf{h}(\dot{\theta}, t) \end{bmatrix},$$

$$\mathbf{h}(\dot{\theta}, t) = \begin{bmatrix} \dot{\theta} v_x(t) \\ 0 \end{bmatrix},$$

$$\mathbf{B}_r = \begin{bmatrix} 0 \\ \mathbf{I} \end{bmatrix},$$

$$\mathbf{g}_s(\theta) = \begin{bmatrix} 0 \\ \mathbf{J} \end{bmatrix},$$

and

$$\mathbf{J} = \begin{bmatrix} -1 & h_1 \\ -1 & h_2 \end{bmatrix},$$

$$\mathbf{R}(\dot{\theta}, \mathbf{s}, \dot{\mathbf{s}}) = -\mathcal{D}^2(\omega) \mathbf{s} - 2\mathcal{D}(\zeta) \mathcal{D}(\omega) \dot{\mathbf{s}} + \mathbf{s} \dot{\theta}^2,$$

where $\mathcal{D}(\omega)$ denotes the diagonal matrix $\text{diag}\{\omega_1, \dots, \omega_n\}$.

Equation (14) can be rewritten as

$$\dot{\mathbf{x}}_r - \mathbf{A}_r \mathbf{x}_r - \mathbf{B}_r \hat{\mathbf{v}} - \mathbf{H}(\dot{\theta}, t) = \mathbf{A}_s \mathbf{x}_s. \quad (16)$$

In designing the reduced-order observer, we consider the left-hand side of (16) to be known quantities. Thus, (16) relates the measurable or known quantities and unmeasurable quantities of the state.

The reduced-order observer equation can now be expressed as

$$\dot{\hat{\mathbf{x}}}_s = \mathbf{f}_s(\dot{\theta}, \hat{\mathbf{x}}_s, t) + \mathbf{g}_s \hat{\mathbf{v}} + \mathbf{L}(\dot{\mathbf{x}}_r - \mathbf{A}_r \mathbf{x}_r - \mathbf{B}_r \hat{\mathbf{v}} - \mathbf{H}(\dot{\theta}, t) - \mathbf{A}_s \hat{\mathbf{x}}_s). \quad (17)$$

Let $\tilde{\mathbf{x}}_s = \mathbf{x}_s - \hat{\mathbf{x}}_s$ denote the observer error. Clearly,

$$\dot{\tilde{\mathbf{x}}}_s = \mathbf{f}_s(\dot{\theta}, \mathbf{x}_s, t) - \mathbf{f}_s(\dot{\theta}, \hat{\mathbf{x}}_s, t) - \mathbf{L} \mathbf{A}_s \tilde{\mathbf{x}}_s. \quad (18)$$

Define

$$\hat{\mathbf{L}} = [L_1, L_2, L_3, L_4]^T,$$

$$\hat{\mathbf{a}} = [a_1, a_2, a_3, a_4]^T.$$

Then

$$\mathbf{L} = [\mathbf{0} \ \mathbf{0} \ \mathbf{0} \ \hat{\mathbf{L}}],$$

$$\mathbf{A}_s^T = [\mathbf{0} \ \mathbf{0} \ \mathbf{0} \ \hat{\mathbf{a}}],$$

and thus $\mathbf{L} \mathbf{A}_s = \hat{\mathbf{L}} \hat{\mathbf{a}}^T$. Now it can be shown that

$$\mathbf{f}_s(\dot{\theta}, \mathbf{x}_s, t) - \mathbf{f}_s(\dot{\theta}, \hat{\mathbf{x}}_s, t) = (\hat{\mathbf{A}} + \hat{\mathbf{B}}(t)) \tilde{\mathbf{x}}_s,$$

where

$$\hat{\mathbf{A}} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ \hat{\mathbf{A}}_1 & \hat{\mathbf{A}}_2 \end{bmatrix},$$

$$\hat{\mathbf{B}}(t) = \dot{\theta}^2(t) \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{I} & \mathbf{0} \end{bmatrix},$$

$$\hat{\mathbf{A}}_1 = \begin{bmatrix} h_1 a_1 - \omega_1^2 & h_1 a_2 \\ h_2 a_1 & h_2 a_2 - \omega_2^2 \end{bmatrix},$$

$$\hat{\mathbf{A}}_2 = \begin{bmatrix} h_1 a_3 - 2\zeta_1 \omega_1 & h_1 a_4 \\ h_2 a_3 & h_2 a_4 - 2\zeta_2 \omega_2 \end{bmatrix}.$$

Therefore observer-error dynamics can be rewritten as

$$\dot{\tilde{\mathbf{x}}}_s = (\hat{\mathbf{A}} - \hat{\mathbf{L}} \hat{\mathbf{a}}^T) \tilde{\mathbf{x}}_s + \hat{\mathbf{B}}(t) \tilde{\mathbf{x}}_s. \quad (19)$$

Clearly, the conditions given in the previous section guarantee that $\hat{\mathbf{B}}(t) \rightarrow 0$ as $t \rightarrow \infty$. Thus, we first choose $\hat{\mathbf{a}}$ such that the pair $(\hat{\mathbf{A}}, \hat{\mathbf{a}}^T)$ is completely observable. Then, we assign the eigenvalues of the matrix $\hat{\mathbf{A}} - \hat{\mathbf{L}} \hat{\mathbf{a}}^T$ to ensure that the error dynamics given by (19) are asymptotically stable and adequately fast.

V. SIMULATION

In this section, we demonstrate the effectiveness of the Lyapunov-based controller (12)-(13) that uses the observed slosh state via (17) by applying the controller to the complete nonlinear system (1)-(4). For simplicity, only the first two slosh modes are included. The physical parameters used in the simulations are given in Table 1. The fluid parameters are obtained using the formulae in [7].

We consider stabilization of the spacecraft in orbital transfer, suppressing the transverse and pitching motion of the spacecraft and sloshing of fuel while the spacecraft is accelerating. In other words, the control objective is to stabilize the relative equilibrium corresponding to a constant axial spacecraft acceleration of 1.72 m/s^2 and $v_z = \theta = \dot{\theta} = s_i = \dot{s}_i = 0$, $i = 1, 2$.

Table I
PHYSICAL PARAMETERS.

Parameter	Value	Parameter	Value
m	975 kg	F	2450 N
I	400 kg · m ²	I_0	14.85 kg · m ²
m_0	358 kg	k_1	750 N/m
m_1	89 kg	k_2	65 N/m
m_2	2.7 kg	c_1	25.8 N · s/m
h_0	-0.011 m	c_2	1.32 N · s/m
h_1	0.035 m	b	-0.6 m
h_2	0.291 m	d	1.2 m

Time responses shown in Figs. 2-4 correspond to the initial conditions $v_{x_0} = 3000$ m/s, $v_{z_0} = 100$ m/s, $\theta_0 = 5^\circ$, $\dot{\theta}_0 = 0$, $s_{1_0} = 0.147$ m, $s_{2_0} = -0.147$ m, and $\dot{s}_{1_0} = \dot{s}_{2_0} = 0$. We assume a fuel burn time of 660 s. As can be seen in the figures, the transverse velocity, attitude angle, and the slosh states converge to the relative equilibrium at zero while the axial velocity v_x increases and $\dot{v}_x$ tends asymptotically to 1.72 m/s².

The parameters of the controller given by (12)-(13) were chosen as $r_1 = 8 \times 10^{-7}$, $r_2 = 2500$, $r_3 = 500$, $r_4 = 1 \times 10^{-5}$, $l_1 = 10000$, $l_2 = 40000$.

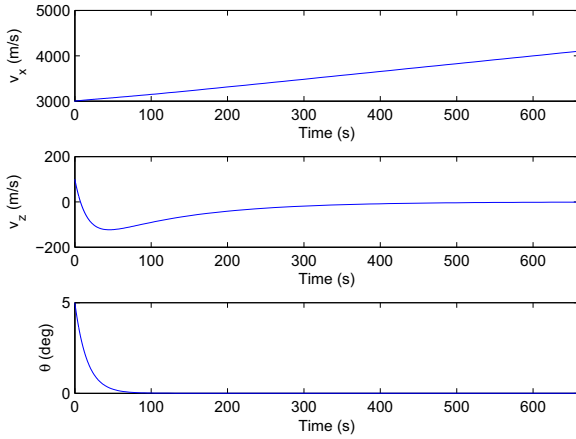


Figure 2. Time responses of v_x , v_z and θ .

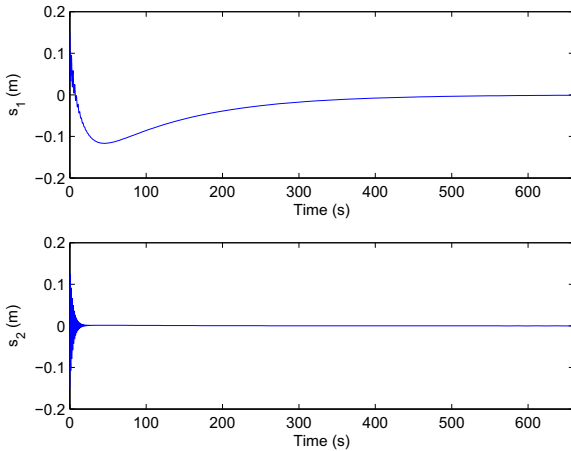


Figure 3. Time responses of s_1 and s_2 .

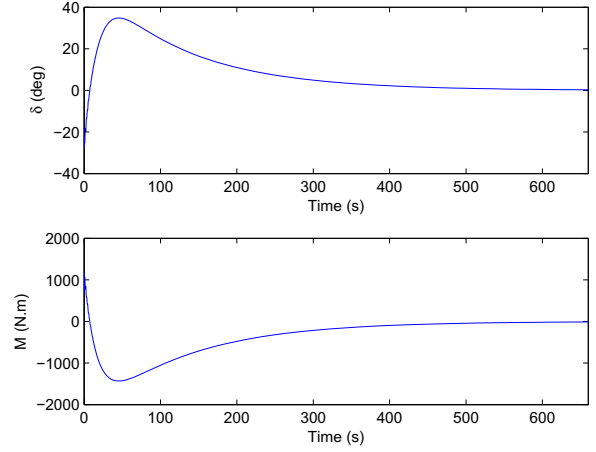


Figure 4. Gimbal deflection angle δ and pitching moment M .

The observer (17) was implemented by placing the eigenvalues of the matrix $\hat{A} - \hat{L}\hat{a}^T$ at $(-4, -5, -6, -7)$ using

$$\hat{a} = [1, 1, 1, 1]^T,$$

$$\hat{L} = [6.2615, -6.8389, 19.7105, 2.4079]^T.$$

VI. CONCLUSIONS

We have considered the problem of observer-based control of slosh in planar maneuvering space vehicles in a zero gravity environment. A nonlinear feedback that uses a reduced-order observer for the estimation of the slosh state has been proposed for the control of the translational velocity vector and the attitude of the spacecraft while suppressing the sloshing modes. The effectiveness of the proposed observer-based control law has been illustrated through a computer simulation.

VII. ACKNOWLEDGMENTS

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